

Homework 9

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Instructions

Answer the following questions and complete the exercises in RMarkdown. Please embed all of your code and push your final work to your repository. Your final lab report should be organized, clean, and run free from errors. Remember, you must remove the `#` for the included code chunks to run. Be sure to add your name to the author header above.

Make sure to use the formatting conventions of RMarkdown to make your report neat and clean!

Load the libraries

```
library(tidyverse)
library(janitor)
```

Data

For this homework, we will use a data set compiled by the Office of Environment and Heritage in New South Wales, Australia. It contains the enterococci counts in water samples obtained from Sydney beaches as part of the Beachwatch Water Quality Program. *Enterococci* are bacteria common in the intestines of mammals; they are rarely present in clean water. So, *Enterococci* values are a measurement of pollution. *cfu* stands for colony forming units and measures the number of viable bacteria in a sample cfu.

This homework loosely follows the tutorial of R Ladies Sydney. If you get stuck, check it out!

1. Start by loading the data `sydneybeaches`. Do some exploratory analysis to get an idea of the data structure.

```
sydneybeaches<- read_csv("data/sydneybeaches.csv")
```

```
## Rows: 3690 Columns: 8
## -- Column specification -----
## Delimiter: ","
## chr (4): Region, Council, Site, Date
## dbl (4): BeachId, Longitude, Latitude, Enterococci (cfu/100ml)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
sydneybeaches
```

```
## # A tibble: 3,690 x 8
##   BeachId Region Council Site Longitude Latitude Date Enterococci (cfu/100~1
##   <dbl> <chr> <chr> <chr> <dbl> <dbl> <chr> <dbl>
## 1      25 Sydney~ Randwi~ Clov~ 151. -33.9 02/0~ 19
## 2      25 Sydney~ Randwi~ Clov~ 151. -33.9 06/0~ 3
## 3      25 Sydney~ Randwi~ Clov~ 151. -33.9 12/0~ 2
## 4      25 Sydney~ Randwi~ Clov~ 151. -33.9 18/0~ 13
## 5      25 Sydney~ Randwi~ Clov~ 151. -33.9 30/0~ 8
## 6      25 Sydney~ Randwi~ Clov~ 151. -33.9 05/0~ 7
## 7      25 Sydney~ Randwi~ Clov~ 151. -33.9 11/0~ 11
## 8      25 Sydney~ Randwi~ Clov~ 151. -33.9 23/0~ 97
## 9      25 Sydney~ Randwi~ Clov~ 151. -33.9 07/0~ 3
## 10     25 Sydney~ Randwi~ Clov~ 151. -33.9 25/0~ 0
## # i 3,680 more rows
## # i abbreviated name: 1: 'Enterococci (cfu/100ml)'
```

2. Are these data “tidy” per the definitions of the tidyverse? How do you know? Are they in wide or long format?

```
#yes, each column describes one variable
#the data is in the wide format
```

3. We are only interested in the variables site, date, and enterococci_cfu_100ml. Make a new object focused on these variables only. Name the object `sydneybeaches_long`

```
sydneybeaches%>%
  select(Site, Date, 'Enterococci (cfu/100ml)')
```

```
## # A tibble: 3,690 x 3
##   Site      Date      'Enterococci (cfu/100ml) '
##   <chr>    <chr>    <dbl>
## 1 Clovelly Beach 02/01/2013 19
## 2 Clovelly Beach 06/01/2013 3
## 3 Clovelly Beach 12/01/2013 2
## 4 Clovelly Beach 18/01/2013 13
## 5 Clovelly Beach 30/01/2013 8
## 6 Clovelly Beach 05/02/2013 7
## 7 Clovelly Beach 11/02/2013 11
## 8 Clovelly Beach 23/02/2013 97
## 9 Clovelly Beach 07/03/2013 3
## 10 Clovelly Beach 25/03/2013 0
## # i 3,680 more rows
```

```
sydneybeaches_long<- sydneybeaches%>%
  select(Site, Date, 'Enterococci (cfu/100ml)')

sydneybeaches_long
```

```
## # A tibble: 3,690 x 3
##   Site      Date      'Enterococci (cfu/100ml)'
##   <chr>      <chr>      <dbl>
## 1 Clovelly Beach 02/01/2013      19
## 2 Clovelly Beach 06/01/2013       3
## 3 Clovelly Beach 12/01/2013       2
## 4 Clovelly Beach 18/01/2013      13
## 5 Clovelly Beach 30/01/2013       8
## 6 Clovelly Beach 05/02/2013       7
## 7 Clovelly Beach 11/02/2013      11
## 8 Clovelly Beach 23/02/2013     97
## 9 Clovelly Beach 07/03/2013       3
## 10 Clovelly Beach 25/03/2013       0
## # i 3,680 more rows
```

4. Pivot the data such that the dates are column names and each beach only appears once (wide format). Name the object `sydneybeaches_wide`

```
sydneybeaches_long %>%
  pivot_wider(names_from = "Date",
              values_from = "Enterococci (cfu/100ml)")
```

```
## # A tibble: 11 x 345
##   Site      '02/01/2013' '06/01/2013' '12/01/2013' '18/01/2013' '30/01/2013'
##   <chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 Clovelly Be~      19          3          2          13          8
## 2 Coogee Beach      15          4          17          18         22
## 3 Gordons Bay~      NA          NA          NA          NA         NA
## 4 Little Bay ~       9          3          72          1         44
## 5 Malabar Bea~       2          4         390         15         13
## 6 Maroubra Be~       1          1          20          2         11
## 7 South Marou~       1          0          33          2         13
## 8 South Marou~      12          2         110         13        100
## 9 Bondi Beach       3          1           2          1           6
## 10 Bronte Beach      4          2          38          3         25
## 11 Tamarama Be~       1          0           7         22         23
## # i 339 more variables: '05/02/2013' <dbl>, '11/02/2013' <dbl>,
## #   '23/02/2013' <dbl>, '07/03/2013' <dbl>, '25/03/2013' <dbl>,
## #   '02/04/2013' <dbl>, '12/04/2013' <dbl>, '18/04/2013' <dbl>,
## #   '24/04/2013' <dbl>, '01/05/2013' <dbl>, '20/05/2013' <dbl>,
## #   '31/05/2013' <dbl>, '06/06/2013' <dbl>, '12/06/2013' <dbl>,
## #   '24/06/2013' <dbl>, '06/07/2013' <dbl>, '18/07/2013' <dbl>,
## #   '24/07/2013' <dbl>, '08/08/2013' <dbl>, '22/08/2013' <dbl>, ...
```

```
sydneybeaches_wide <- sydneybeaches_long %>%
  pivot_wider(names_from = "Date",
              values_from = "Enterococci (cfu/100ml)")
```

5. Pivot the data back so that the dates are data and not column names.

```
sydneybeaches_wide%>%
  pivot_longer(-Site,
    names_to = "dates",
    values_to = "Enterococci (cfu/100ml)")
```

```
## # A tibble: 3,784 x 3
##   Site      dates      'Enterococci (cfu/100ml)'
##   <chr>      <chr>      <dbl>
## 1 Clovelly Beach 02/01/2013      19
## 2 Clovelly Beach 06/01/2013       3
## 3 Clovelly Beach 12/01/2013       2
## 4 Clovelly Beach 18/01/2013      13
## 5 Clovelly Beach 30/01/2013       8
## 6 Clovelly Beach 05/02/2013       7
## 7 Clovelly Beach 11/02/2013      11
## 8 Clovelly Beach 23/02/2013     97
## 9 Clovelly Beach 07/03/2013       3
## 10 Clovelly Beach 25/03/2013       0
## # i 3,774 more rows
```

6. We haven't dealt much with dates yet, but separate the date into columns day, month, and year. Do this on the `sydneybeaches_long` data.

```
sydneybeaches_long%>%
  separate(Date, into=c("day", "month", "year"))
```

```
## # A tibble: 3,690 x 5
##   Site      day month year 'Enterococci (cfu/100ml)'
##   <chr>      <chr> <chr> <chr>      <dbl>
## 1 Clovelly Beach 02    01   2013      19
## 2 Clovelly Beach 06    01   2013       3
## 3 Clovelly Beach 12    01   2013       2
## 4 Clovelly Beach 18    01   2013      13
## 5 Clovelly Beach 30    01   2013       8
## 6 Clovelly Beach 05    02   2013       7
## 7 Clovelly Beach 11    02   2013      11
## 8 Clovelly Beach 23    02   2013     97
## 9 Clovelly Beach 07    03   2013       3
## 10 Clovelly Beach 25    03   2013       0
## # i 3,680 more rows
```

7. What is the average `enterococci_cfu_100ml` by year for each beach. Think about which data you will use- long or wide.

```
sydneybeaches_new<-sydneybeaches_long %>%
  separate(Date, into=c("day", "month", "year"))%>%
  select(site=Site, day, month, year, contamination="Enterococci (cfu/100ml)")

sydneybeaches_new%>%
  group_by(site, year)%>%
  summarize(mean=mean(contamination, na.rm=T))
```

```
## 'summarise()' has grouped output by 'site'. You can override using the
## '.groups' argument.
```

```
## # A tibble: 66 x 3
## # Groups:   site [11]
##   site      year  mean
##   <chr>    <chr> <dbl>
## 1 Bondi Beach 2013   32.2
## 2 Bondi Beach 2014   11.1
## 3 Bondi Beach 2015   14.3
## 4 Bondi Beach 2016   19.4
## 5 Bondi Beach 2017   13.2
## 6 Bondi Beach 2018   22.9
## 7 Bronte Beach 2013   26.8
## 8 Bronte Beach 2014   17.5
## 9 Bronte Beach 2015   23.6
## 10 Bronte Beach 2016   61.3
## # i 56 more rows
```

8. Make the output from question 7 easier to read by pivoting it to wide format.

```
sydneybeaches_new_contamination_by_year<-sydneybeaches_new%>%
  group_by(site, year)%>%
  summarize(mean=mean(contamination, na.rm=T))
```

```
## 'summarise()' has grouped output by 'site'. You can override using the
## '.groups' argument.
```

```
sydneybeaches_new_contamination_by_year%>%
  pivot_wider(names_from = "site",
              values_from = "mean")
```

```
## # A tibble: 6 x 12
##   year 'Bondi Beach' 'Bronte Beach' 'Clovelly Beach' 'Coogee Beach'
##   <chr>    <dbl>         <dbl>         <dbl>         <dbl>
## 1 2013      32.2          26.8          9.28          39.7
## 2 2014      11.1          17.5          13.8          52.6
## 3 2015      14.3          23.6          8.82          40.3
## 4 2016      19.4          61.3          11.3          59.5
## 5 2017      13.2          16.8          7.93          20.7
## 6 2018      22.9          43.4          10.6          21.6
## # i 7 more variables: 'Gordons Bay (East)' <dbl>, 'Little Bay Beach' <dbl>,
## #   'Malabar Beach' <dbl>, 'Maroubra Beach' <dbl>,
## #   'South Maroubra Beach' <dbl>, 'South Maroubra Rockpool' <dbl>,
## #   'Tamarama Beach' <dbl>
```

9. What was the most polluted beach in 2013?

```
sydneybeaches_new_contamination_by_year%>%
  filter(year=="2013")%>%
  arrange(desc(mean))
```

```
## # A tibble: 11 x 3
## # Groups:   site [11]
##   site          year    mean
##   <chr>         <chr> <dbl>
## 1 Little Bay Beach    2013  122.
## 2 Malabar Beach      2013  101.
## 3 South Maroubra Rockpool 2013   96.4
## 4 Maroubra Beach     2013   47.1
## 5 Coogee Beach       2013   39.7
## 6 South Maroubra Beach 2013   39.3
## 7 Bondi Beach        2013   32.2
## 8 Tamarama Beach     2013   29.7
## 9 Bronte Beach       2013   26.8
## 10 Gordons Bay (East) 2013   24.8
## 11 Clovelly Beach    2013    9.28
```

#Little Bay Beach was the most polluted beach in 2013 with a mean contamination value of around 121 cfu

```
?mutate
```

```
## starting httpd help server ... done
```

10. Explore the data! Do one analysis of your choice that includes a minimum of three lines of code.

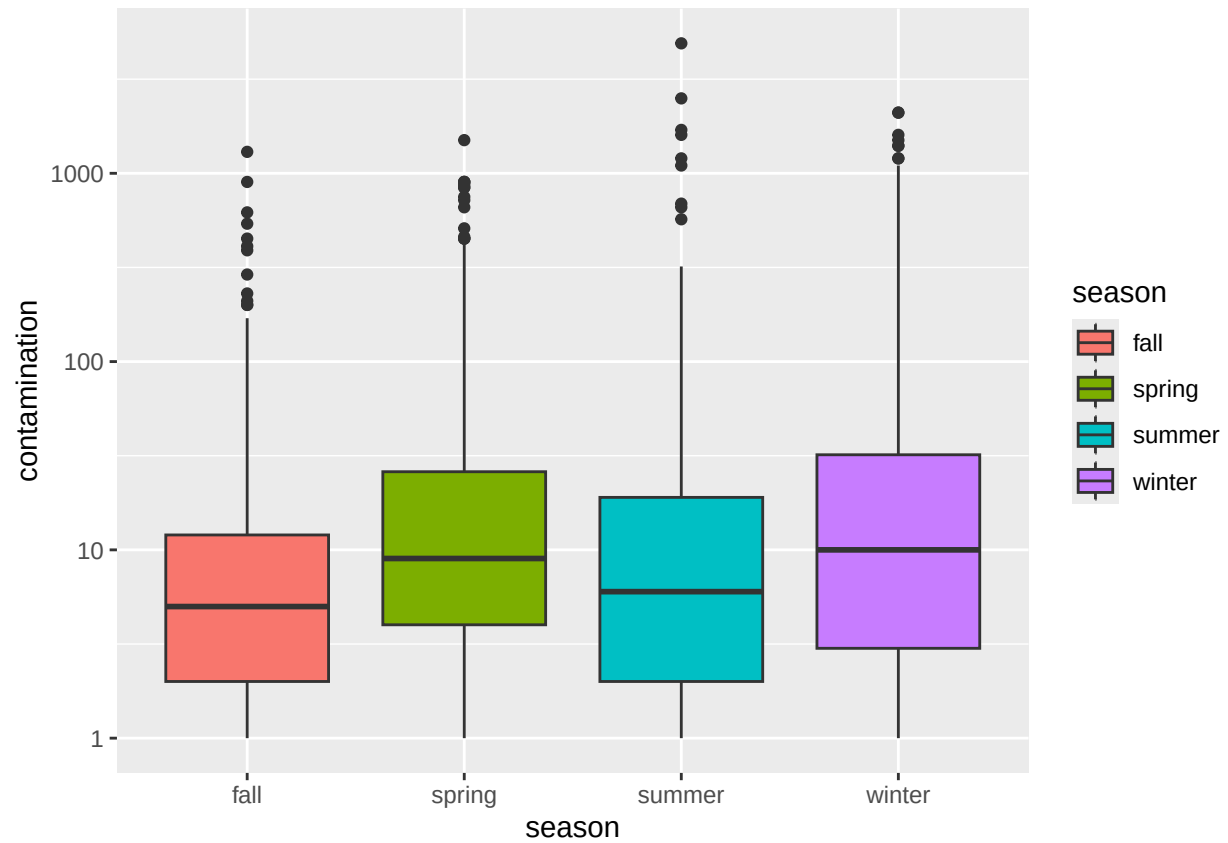
#Is there a higher rate of contamination in the summer compared to the other months?

#mutate to define summer

```
sydneybeaches_new%>%
  #separate(date, into=c("day", "month", "year"), sep="/")%>%
  mutate(season=case_when(month=="12" | month=="01" | month=="02"~ "winter",
                           month=="03" | month=="04" | month=="05" ~ "spring",
                           month=="06" | month=="07" | month=="08" ~ "summer",
                           month=="09" | month=="10" | month=="11"~ "fall"))%>%
  select(season, contamination)%>%
  #group_by(season)%>%
  #summarise(mean_cont=mean(contamination, na.rm=T))%>%
  ggplot(aes(x=season, y=contamination, fill=season))+
  geom_boxplot()+
  scale_y_log10()
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 616 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



Knit and Upload

Please knit your work as an .html file and upload to Canvas. Homework is due before the start of the next lab. No late work is accepted. Make sure to use the formatting conventions of RMarkdown to make your report neat and clean!